Preference based Contrastive Reinforcement Learning

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***Abstract*—Preference-based Reinforcement Learning (PbRL) is a promising avenue for reward specification, yet it faces significant hurdles when preference labels are granular (state-wise) and noisy, typical of data from non-expert humans or Vision-Language Models. In this work, we present a robust framework for learning from such noisy, state-wise preferences by leveraging the environment's transition dynamics as a ground-truth regularizer. Building upon Contrastive Goal-Conditioned RL (C-GCRL), our method interleaves offline updates from the noisy preference dataset with online updates grounded in physical reachability. We hypothesize that while preference data directs exploration towards high-value regions, the dense, physics-consistent signal from online interactions can override spurious local maxima caused by label noise. We introduce data augmentation strategies—specifically "Noisy Expert" data generation and preference-guided subgoal sampling—that are essential for effective learning in this sparse regime. Evaluations on the MetaWorld benchmark demonstrate that our approach performs comparably to baselines like C-GCRL and Inverse Preference Learning (IPL), successfully recovering working policies even in the presence of substantial preference noise.**

I. Introduction & Motivation

Specifying reward functions remains one of the most significant bottlenecks in deploying Reinforcement Learning (RL) to real-world robotic systems. While sparse rewards (binary success/failure) are easy to specify, they make exploration intractable. Conversely, dense, shaped rewards steer exploration but are prone to "reward hacking" and require deep domain expertise to tune.

Preference-based RL (PbRL) has emerged as a promising alternative, replacing manual reward engineering with binary comparisons (e.g., "Segment A is better than Segment B"). However, scaling PbRL to the level required for generalist robots presents a two-fold challenge that remains unsolved in current literature:

1) **Granularity vs. Sparsity:** While trajectory-based comparisons are rich in information, they result in the "credit assignment problem"—it is unclear which specific state in the trajectory caused the high rating. *State-wise* preferences (comparing state s_1 vs s_2) solve credit assignment but are extremely sparse. A single bit of information ($s_1 \succ s_2$) tells the agent nothing about the absolute value of the states, the temporal distance to the goal, or the feasibility of the transition.
2) **The Destructive Nature of Noise:** As we scale data collection using crowdsourcing or automated proxy annotators (such as GPT-4V or CLIP), noise becomes inevitable. In standard supervised learning, random label noise often averages out. In preference learning, however, noise is structural. A flipped preference label ($s_{bad} \succ s_{good}$) creates a "poisoned" reward signal that actively encourages the agent to seek failure states or unsafe regions.

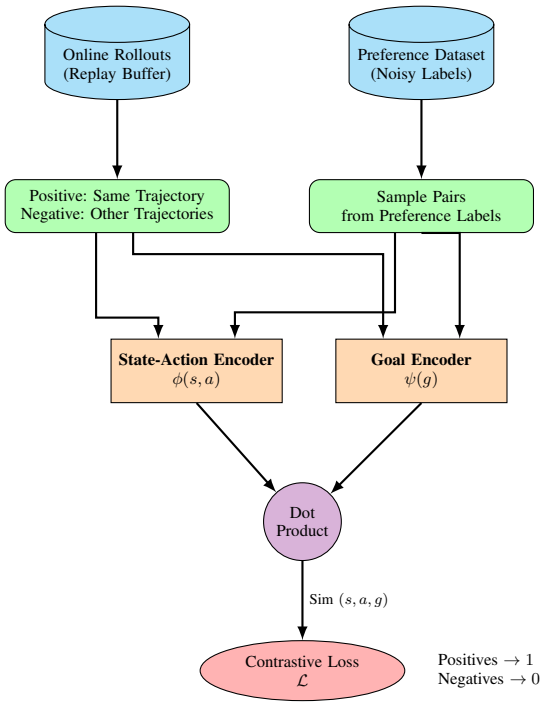


Fig. 1. **Architecture Overview.** We sample contrastive pairs from both Online Rollouts and the Preference Dataset. For online rollouts, positive samples are drawn from the same trajectory (future states) and negative samples from other trajectories in the batch. These pairs are processed by a *State-Action Encoder* and a *Goal Encoder*. The dot product of their outputs represents the reachability/value score. The loss function pushes the score of positive pairs towards 1 and negative pairs towards 0, effectively structuring the latent space according to both dynamics and preferences.

Learning from noisy, state-wise preferences presents a formidable challenge in robust reward inference. The agent is tasked with simultaneously inferring the reward function and the optimal policy from sparse noisy data. Current methods like PEBBLE [23] or SURF [28] primarily address noise in trajectory segments through filtering or re-labeling techniques. However, these approaches often treat the reward function as an abstract classifier, neglecting the fundamental physical constraints of the environment. Consequently, they lack a

mechanism to ground these preferences in physical reality, making them susceptible to learning policies that are locally optimal with respect to noisy labels but dynamically infeasible or suboptimal.

Contribution. We propose a novel framework that leverages environment interactions via *Contrastive Learning* to address the intractability of noisy preference labels. Prior work in hybrid reinforcement learning has shown that combining offline data with online interactions can lead to more statistically and computationally efficient learning by mitigating both the sample inefficiency of pure online RL and the distributional challenges of pure offline RL[34]. Building on this finding, we hypothesize that the environment's underlying dynamics can serve as a potent regularizer against label noise. To illustrate, consider a scenario where a noisy preference label incorrectly asserts that state A is superior to state B ($A \succ B$), whereas state B is, in reality, the superior state closer to the goal. In a standard PbRL setting, the agent would be incentivized to maximize the value of A. However, by anchoring the preference learning process to a Contrastive Goal-Conditioned RL (C-GCRL) backbone, we enforce *dynamical consistency*. Since state B is physically closer to the goal distribution, the contrastive reachability objective will inherently assign a higher value to B (representing higher reachability probability). This dense, physics-informed signal acts as a strong prior that overrides the misleading gradient from noisy preference labels. Thus, our method accounts for the reachability structure of the environment, even in the presence of contradictory preference data.

II. Related Work

Preference-Based Reinforcement Learning. Preference-based reinforcement learning (PbRL) replaces explicit scalar rewards with pairwise comparisons or rankings that indicate which trajectory segment is preferred [9]. Early PbRL algorithms [2, 3] learned low-dimensional reward functions from limited human feedback. With deep networks and large-scale feedback collection, PbRL has been extended to continuous control domains [7, 17, 22, 13, 33, 6]. These approaches typically follow a two-stage procedure: first train a reward model from pairwise preferences using a Bradley–Terry likelihood, then train a policy via standard RL on that learned reward. Recent work has explored end-to-end formulations that eliminate the intermediate reward model and learn directly from preferences.[14, 15]. Our work departs from prior preference-based RL approaches along two key dimensions. First, we explicitly model and learn under noisy preference labels, whereas existing methods typically assume clean or human-perfect comparisons. Second, instead of backpropagating preferences directly through the policy/reward model/value function, we embed them within a goal-conditioned contrastive RL formulation, where a contrastive Q-function representations integrate preferences into the value structure itself. This allows the offline Q-function learning to be structurally similar to the online Q-function learning of GCRL.

Goal-conditioned reinforcement learning (GCRL). Goal-conditioned reinforcement learning (GCRL) can be viewed as a special case of multi-task RL, where each task is defined by a desired goal state g that the agent aims to reach [18, 32, 26, 10]. From a preference-based perspective, GCRL can be interpreted as optimizing a single global preference—the goal state being preferred over all others—while auxiliary objectives such as temporal distance provide additional shaping signals for learning. Our work builds upon contrastive formulations of GCRL [8], where the Q-function is represented as a similarity between state–action and goal embeddings. We extend this framework by integrating preference information directly into the contrastive objective, enabling the agent to leverage preferences as directional signals that guide exploration and reduce sensitivity to noisy feedback during online policy learning.

Offline-to-online reinforcement learning. While online RL algorithms are powerful, they typically require a large number of environment interactions to learn effective policies from scratch [19, 37, 12]. To mitigate this sample inefficiency, a growing body of work leverages *offline data* to accelerate online learning. Existing approaches integrate offline data in several ways: by seeding the replay buffer with demonstration or expert trajectories [31, 36, 16, 34], incorporating auxiliary behavior cloning or imitation losses during policy optimization [30, 20, 38, 39], or distilling high-level skills and latent representations for subsequent online fine-tuning [11, 1]. A different line of work first runs offline RL for learning a good policy and value initialization from the offline data, followed by online fine-tuning [24, 27, 23, 21, 25] However, nearly all such methods assume access to full trajectories with reward annotations. The recent HyPO algorithm [35] is an exception, leveraging offline *preferences* to bootstrap online policy improvement, though it has been explored primarily in large language model settings where they start with a good reference policy. In contrast, our work studies the offline-to-online transition given preference data, where only pairwise comparisons rather than full trajectories are available.

III. Problem Formulation

We model the agent's environment as a Goal-Conditioned Markov Decision Process (GCMDP), defined by the tuple $\mathcal{M} = (S, A, G, T, R, \gamma)$, where:

- S is the set of states.
- A is the set of actions.
- $G \subseteq S$ is the set of possible goals.
- $T(s'|s, a)$ is the unknown state transition function.
- $R(s, g)$ is the true, underlying reward function, which is unknown to the agent.
- $\gamma \in [0, 1)$ is the discount factor.

In addition, the agent is provided with a dataset that consists of N tuples, $D_n = \{(s_i^1, a_i^1, s_i^2, a_i^2, \mu_i)\}_{i=1}^N$, where $s_i^1, s_i^2 \epsilon S$, and μ_i is the noisy preference label, such that

$$P(\mu_i = 1) = \sigma(R(s_i^1, a_i^1, g) - R(s_i^2, a_i^2, g) + \varepsilon_i)$$

where ε_i is the noise term. We assume these preferences are generated by a Boltzmann Rational [4] labeler with respect to a single, implicit task goal g_{task}.

A. Project Scope and Objectives

The agent's objective is to learn a goal-conditioned policy $\pi(a \mid s, g)$ that maximizes the expected discounted return

$$J(\pi) = \mathbb{E}_{\pi,T,g\sim G}\left[\sum_{t=0}^{T} \gamma^t R(s_t, \pi(a_t), g)\right],$$

using only noisy preference feedback and environment interaction, without access to the true reward function R subject to the constraint that we only have access to the noisy dataset $\mathcal{D}_{pref}$.

The scope is strictly limited to **state-wise preferences**(segment length=1). We assume the noise rate η is constant but unknown to the agent. The core technical objective is to demonstrate that grounding the preference learning in the *reachability dynamics* of the GC-MDP (via contrastive learning) provides a robust regularization against the noise term η. We evaluate performance through the task success rate of the trained policy.

IV. Proposed Approach

Our approach integrates the noisy preference signal directly into a goal-conditioned policy learning framework. We construct a method that interleaves online exploration guided by preference data with a dual-stream contrastive update mechanism.

A. Background: Contrastive Goal-Conditioned RL

We build on the Contrastive Goal-Conditioned Reinforcement Learning (C-GCRL) method [8], which represents the goal-conditioned Q-function as an inner product between state–action and goal embeddings:

$$Q_\theta(s, a, g) = \phi_\theta(s, a)^\top \psi_\theta(g),$$

where ϕ_θ and ψ_θ denote learnable encoders for state–action pairs and goals, respectively. C-GCRL learns these embeddings via a binary contrastive objective that encourages high similarity for matching pairs (valid transitions) and low similarity for mismatched ones:

$$\mathcal{L}_{\text{C-GCRL}}(\theta) = -\mathbb{E}_{(s,a,s'_f,s_f^-)}\Big[\log \sigma\big(\phi_\theta(s, a)^\top \psi_\theta(s'_f)\big) + \log\big(1 - \sigma(\phi_\theta(s, a)^\top \psi_\theta(s_f^-))\big)\Big]$$

where s'_f is a future state reachable from (s, a), and s_f^- is a random (unreachable) state. This objective implicitly optimizes a contrastive form of the Bellman equation, aligning embeddings with the environment's dynamics.

Algorithm 1 Contrastive RL with Preference Learning

Require: Preference Dataset $\mathcal{D}_{\text{pref}}$, Online Buffer $\mathcal{B}_{\text{on}}$, Steps T, Collection Interval K, UTD Ratio U
Ensure: Policy π_θ, Critic Q_ϕ, $\mathcal{B}_{\text{on}} \leftarrow \emptyset$

for step $t = 1$ **to** T **do**
  // 1. Goal-Conditioned Online Collection
  if $t \bmod K = 0$ **then**
    Sample goal $g \sim \mathcal{D}^+_{\text{pref}}$
    // Sample from positive preference states
    Collect τ via $\pi_\theta(\cdot|s, g)$ w/ explore noise
    $\mathcal{B}_{\text{on}} \leftarrow \mathcal{B}_{\text{on}} \cup \{\tau\}$
  end if
  // 2. Offline Preference Update
  Sample batch $b_{\text{pref}} \sim \mathcal{D}_{\text{pref}}$
  Update representations $\phi(s, a)$, $\psi(g)$ and policy $\pi(a|s, g)$ using Pref. Contrastive RL
  // 3. Online Update w/ Update Ratio
  for $j = 1$ **to** U **do**
    Sample batch $b_{\text{on}} \sim \mathcal{B}_{\text{on}}$
    Update representations $\phi(s, a)$, $\psi(g)$ and policy $\pi(a|s, g)$ using Std. Contrastive RL
  end for
end for

B. Preference-Guided Exploration

A key challenge in sparse reward settings is collecting relevant experience. Standard exploration often fails to visit the regions of state space necessary to disambiguate noisy preferences. To address this, we leverage the positive samples from our preference dataset $\mathcal{D}_{pref}$ as an inductive bias for exploration.

We define an *exploration goal distribution* formed by the set of all preferred states in the dataset: $G_{exp} = \{s^+ \mid (s^+, a^+, s^-, a^-) \in \mathcal{D}_{pref}\}$. During data collection, we sample a goal $g_{explore} \sim G_{exp}$ and execute the current policy $\pi(s, g_{explore})$ with exploration noise.

$$a_t \sim \pi(\cdot|s_t, g_{explore}) + \mathcal{N}(0, \Sigma)$$

The intuition is that s^+ states, being preferred, represent useful subgoals or waypoints closer to the true task goal. Directing the agent towards these states yields a richer online replay buffer $\mathcal{B}_{online}$ containing trajectories that are task-relevant, thereby facilitating better dynamics learning around high-value regions.

C. Training

We propose a hybrid training loop that alternates between learning from static preferences (Offline) and learning from environment dynamics (Online). The task goals g_{task} used for training are sampled from the distribution of final states of the trajectories used to generate $\mathcal{D}_{pref}$. Refer to Figure 1 for an overview of the entire algorithm.

1. Offline Preference Updates (The Inductive Bias): We sample batches of preference pairs (s^+, a^+, s^-, a^-) from $\mathcal{D}_{pref}$. We treat these preferences as a contrastive signal relative to the task goal g_{task}. We formulate the preference loss by treating the preferred tuple (s^+, a^+) as a "positive"

sample (similar to a reachable future state) and (s^-, a^-) as a "negative" sample.

$$\mathcal{L}_{pref} = -\mathbb{E}_{\mathcal{D}_{pref}}\Big[\log \sigma(\phi_\theta(s^+, a^+)^\top \psi_\theta(g_{task})) + \log(1 - \sigma(\phi_\theta(s^-, a^-)^\top \psi_\theta(g_{task})))\Big]$$

This loss explicitly shapes the embedding space such that preferred states are geometrically closer to the goal embedding than non-preferred states.

2. Online Dynamics Updates (The Regularizer): We sample transitions (s, a, s') from the online replay buffer $\mathcal{B}_{online}$. We apply the standard C-GCRL loss $\mathcal{L}_{\text{C-GCRL}}$ to these transitions. This step ensures that the learned Q-values respect physical reachability. If a noisy preference implies a jump between disconnected states, this dynamics loss will penalize it, as s' (the physically resulting state) will not match the hallucinated high-value state.

3. The Update Ratio U: To balance these two signals, we perform U steps of online dynamics updates for every single step of offline preference update.

$$\frac{\text{Steps}(\mathcal{L}_{\text{C-GCRL}})}{\text{Steps}(\mathcal{L}_{pref})} = U$$

This ratio U is a critical hyperparameter as shown in the experiments section. It is typically set to $U \geq 1$. The rationale is twofold. First, while the offline preference updates provide crucial initial guidance and direct exploration towards useful regions of the state space, they are derived from a noisy dataset which would otherwise lead to a suboptimal policy. Second, although the online signals may initially be noisy due to the suboptimality of the exploration policy, they are strictly grounded in the environment's physics. As training progresses, the online updates become increasingly accurate and serve as a correction mechanism. For states where the preference dataset provides incorrect (noisy) labels, the online updates—governed by true environmental dynamics—will naturally oppose and override these hallucinations, thereby fixing the policy and ensuring robustness.

V. Results

We evaluated our method on the **MetaWorld** benchmark, specifically focusing on the *Drawer-Open* task. Our experiments were designed to answer three questions: (1) How crucial is the data collection strategy for coverage? (2) Does the online-to-offline update ratio U effectively balance guidance and robustness? (3) How does our method compare to state-of-the-art baselines like IPL in noisy settings? (4) Does learning a separate preference critic help learning?

A. Experimental Setup and Data Collection

Baselines. We compared our approach against three distinct baselines:

1) **Plain C-GCRL:** The standard Contrastive Goal-Conditioned RL algorithm [8] training purely on online data without any preference supervision.

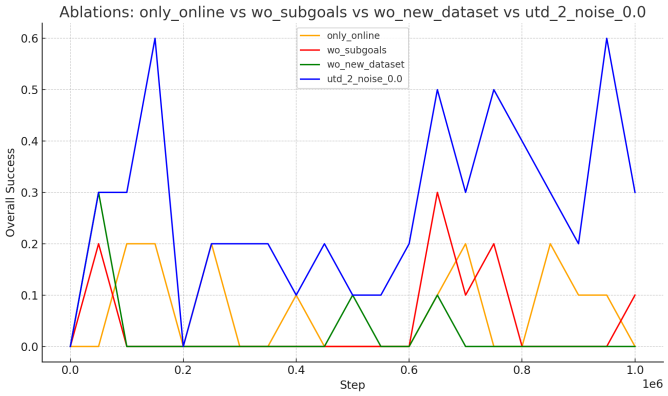


Fig. 2. Ablations for sub-goal sampling, only-online CRL and dataset generation strategy

2) **Direct Preference Optimization (DPO):** An adaptation of DPO for state-wise preferences, which optimizes the policy directly from the static dataset without an explicit reward model[29].
3) **Inverse Preference Learning (IPL):** A PbRL framework that uses the Bradley Terry model and Bellman equations to directly learn a Q-function from preferences [14].

Data Generation Strategy. A critical finding during our preliminary experiments was that standard random data collection failed to provide sufficient state coverage for meaningful preference learning. To address this, we devised a **"Noisy Expert" augmentation strategy**. We utilize an expert policy with $\epsilon = 0.2$ random noise. In 50% of the data collection episodes, the agent acts as this noisy expert throughout the episode. In the other 50% of episodes, the agent toggles to a completely random policy at a uniformly sampled timestep t_{switch}. This strategy ensures a dataset that contains both successful trajectories and diverse failure modes (drift from success), which proved essential for training robust discriminators. We then uniformly sample transitions across these trajectories and determine preferences using privileged information from the simulator (such as the distance of the end-effector to the handle, the distance of the handle from the goal position, etc.). For all of our experiments, we use images as states. The results are shown in Figure 2.

B. Ablation Studies

Impact of Update Ratio (U). We investigated the sensitivity of our method to the ratio of online dynamics updates to offline preference updates, testing $U \in \{1, 2, 5\}$. This hyperparameter governs the trade-off between the inductive bias from preferences and the physical grounding from the environment. We found that $U > 1$ is necessary to suppress noise; specifically, higher ratios allow the dense, physics-consistent online signal to override incorrect preference labels. This implies that higher the UTD the lesser the algorithm relies on the offline preference data. We empirically find that a UTD of 2 performs the best. The results are shown in Figure 3.

Effect of Subgoal Sampling. We ablated the contribution of using positive preference samples as exploration goals

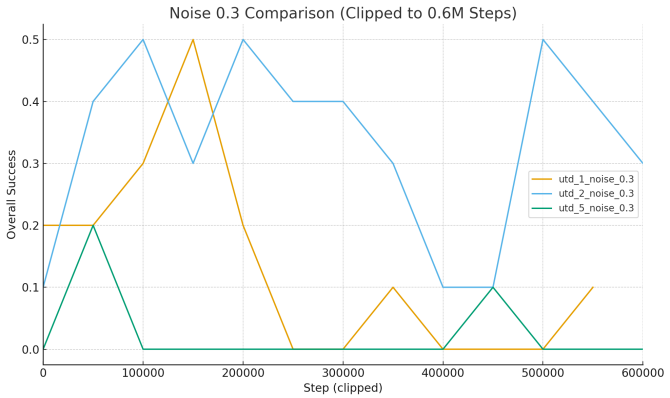


Fig. 3. Ablations for different UTDs

($g_{explore} \sim G_{exp}$). Experiments "without" this mechanism relied on sampling trajectories using the final goal. Results demonstrated that preference-guided exploration is vital; without it, the agent struggles to revisit high-value regions, starving the online buffer of useful transitions and leading to poor convergence. The results are shown in Figure 2.

C. Comparative Analysis

Performance vs. Baselines. We evaluated all methods under varying noise conditions. Both DPO and the base CRL approaches struggled significantly; DPO prone to overfitting the noisy labels directly. Plain CRL failed to solve the task efficiently due to the lack of directed exploration. IPL seems to work well. Our method consistently outperformed these two baselines and matches the performance of IPL.V 4.

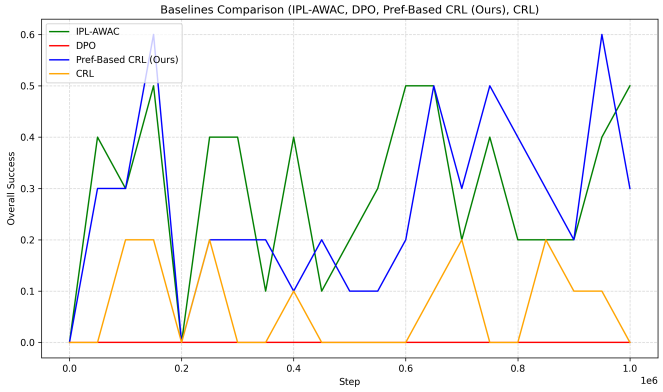


Fig. 4. Comparison of our method - Pref-based CRL with other baselines

Dataset Size Scaling. We examined the sensitivity of our approach to the size of the preference dataset $|\mathcal{D}_{pref}|$. We evaluated performance across dataset sizes $|\mathcal{D}_{pref}| \in \{1000, 2000, 3000, 4000\}$. As anticipated, increasing the volume of preference data correlates with improved policy performance, as the preference signal becomes denser. Figures 5 and 6 illustrate the learning curves under noise levels $\eta = 0.3$ and $\eta = 0.5$, respectively. These results confirm that while larger datasets yield better asymptotic performance, our method maintains robust learning even in lower-data regimes. We present an ablation study of IPL with varying dataset sizes under a noise level of 0.3 in Appendix Figure 12.

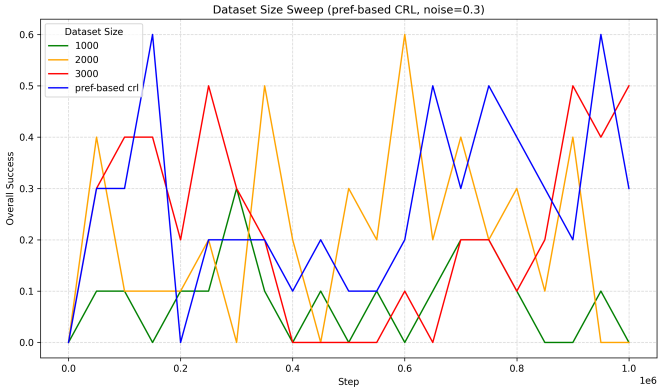


Fig. 5. Ablation with different dataset sizes with a 0.3 noise level

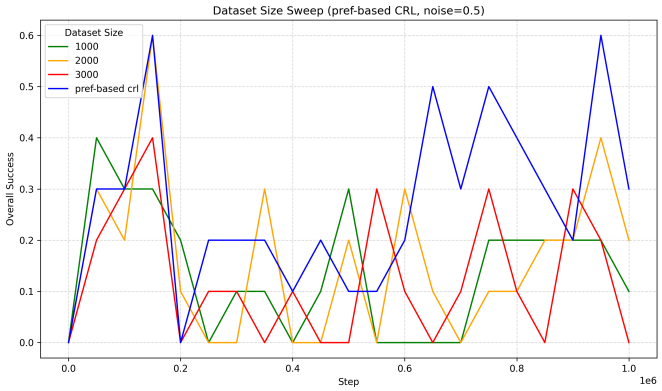


Fig. 6. Ablation with different dataset sizes with a 0.5 noise level

Different noise levels To empirically validate our central hypothesis, that dynamical grounding confers robustness against preference noise we evaluated the framework across a spectrum of noise intensities. We utilized the optimal hyperparameters identified in our preliminary experiments (Subgoal Sampling with an Update-To-Data (UTD) ratio of 2). As illustrated in Figure 7, our method demonstrates comparable resilience to increasing noise levels, maintaining competitive performance. We present an ablation study of IPL under different noise levels in Appendix Figure 13.

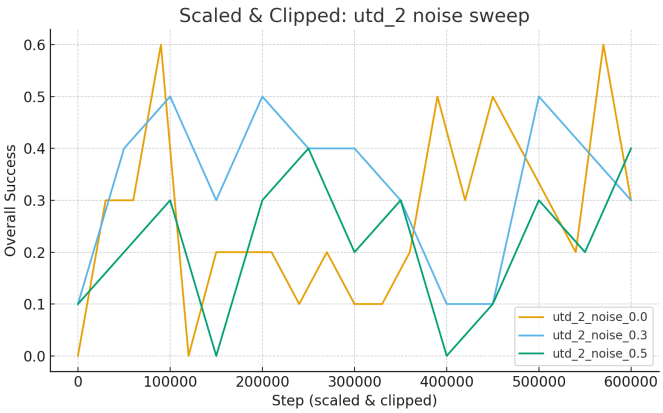


Fig. 7. Ablation on noise levels with UTD 2

Separate Critics We investigated an architectural variant employing *decoupled critics*: one trained exclusively on the

fixed, offline preference dataset, and another learned from online interaction. The final value estimate was constructed as a weighted composition of these two critics, designed to transition from the offline prior to the online estimate over time.

However, as shown in Figure 8, this approach initially had very high performance, but then returns diminished over time. We attribute this to a severe *distribution shift*. The offline critic, trained on a fixed dataset, generalizes poorly to the evolving state distribution visited by the online policy. Consequently, it provides inaccurate guidance during the later phases of learning. Simultaneously, the online critic fails to learn rapidly enough to compensate for the misleading signals from the offline critic.

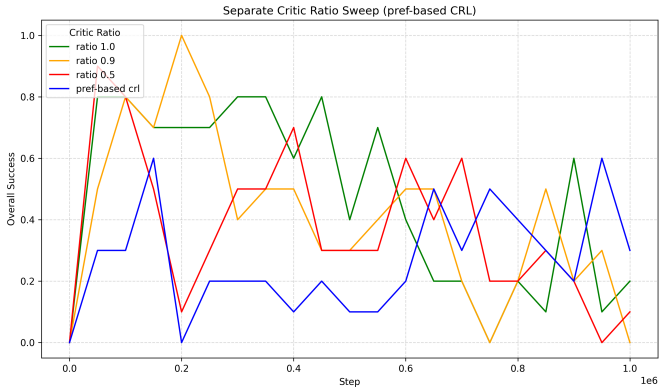


Fig. 8. Ablation on seperate critics with multiple critic ratios

VI. Conclusion

The core challenge addressed in this work is the brittleness of Preference-based RL (PbRL) when dealing with granular, state-wise comparisons corrupted by noise. Solving this problem is of high utility to the robotics community, as robustly learning from state-wise, noisy labels drastically lowers the barrier for supervision.

In our framework, we utilized preferences not as a ground-truth oracle, but as a directional bias to guide exploration and boost the efficiency of the online learner. The integration of online learning proved critical: by grounding value estimation in Contrastive Goal-Conditioned RL (C-GCRL), we essentially used the environment's physics as a regularizer. This dynamics-aware backbone prevents the policy from overfitting to "hallucinated" high-value states found in the noisy dataset, making the method resilient to suboptimal preference labels.

Furthermore, our ablation studies highlighted that algorithmic structure alone is insufficient without careful consideration of the data distribution. We demonstrated that our "Noisy Expert" data generation strategy and "Subgoal Sampling" mechanism were not merely additive, but essential for success. Standard random policies failed to cover goal-relevant regions, while pure expert policies failed to illustrate failure modes; our hybrid strategy bridged this gap. Similarly, given the extreme sparsity of state-wise preference signals, explicit subgoal sampling was required to translate binary bits of information into concrete exploration targets.

The key takeaway is that in the absence of a ground-truth reward signal, the environment's transition dynamics provide a reliable source of supervision. Even when faced with extremely sparse and noisy preference labels, we showed that a policy can recover successful behavior if it anchors its learning in the consistency of physical transitions.

Furthermore, our results underscore the difficulty of bridging offline and online learning when the underlying data distributions are not aligned. In preference-based CRL, we adopt a strategy similar in spirit to Ball et al. [5]: offline preference data is used to bootstrap the agent by warm-starting learning, after which it is interleaved with online rollouts in the replay buffer. In our separate-critic variant, we go one step further and learn two distinct Q-functions: an offline critic trained exclusively on offline preference data, and a second critic trained purely on online interactions. Policy updates then attempt to compose these critics during action selection.

However, as training progresses, we observe that this decomposition leads to instability—both critics drift, ultimately degrading policy performance. These findings suggest that naïvely combining offline and online signals, especially when they originate from heterogeneous distributions or supervision modalities, is fundamentally brittle. Standard design choices that perform well in single-distribution settings do not translate into effective policies at test time, highlighting the need for more principled mechanisms for integrating offline preference data with online exploration.

A. Future Work and Limitations

We acknowledge several limitations due to computational constraints in the scope of this project. Our current evaluation was limited to one random seeds on a single task class. Immediate future work involves scaling this evaluation to a broader suite of MetaWorld tasks and increasing the statistical rigor with more seeds. More broadly, our results highlight that combining offline preference data with online dynamics data requires careful design decisions. The separate-critic ablation illustrates how quickly performance can degrade when the offline preference distribution diverges from the evolving online state distribution. A natural next step is to design mechanisms that adaptively rebalance offline and online updates based on distributional drift. Beyond empirical scaling, we see exciting opportunities in replacing the synthetic noise model with real-world noisy annotators, such as Vision-Language Models (VLMs) or crowd-sourced human feedback. Finally, our framework hints at a natural Active Learning criterion: states where the preference loss and dynamics loss strongly disagree could serve as high-value queries for requesting human intervention, potentially correcting the dataset online.

B. Contributions of Each Team Member

Sreyas: Formulated the approach for Pref-based CRL and separate critic, coded IPL/DPO baseline, ran experiments.
Soumojit: Formulated the approach for Pref-based CRL and the ablations. Lead the writing of the report.
Equal:Coded up our method.

References

[1] Anurag Ajay, Aviral Kumar, Pulkit Agrawal, Sergey Levine, and Ofir Nachum. Opal: Offline primitive discovery for accelerating offline reinforcement learning, 2021. URL https://arxiv.org/abs/2010.13611.

[2] Riad Akrour, Marc Schoenauer, and Michele Sebag. Preference-based policy learning. In Dimitrios Gunopulos, Thomas Hofmann, Donato Malerba, and Michalis Vazirgiannis, editors, *Machine Learning and Knowledge Discovery in Databases*, pages 12–27, Berlin, Heidelberg, 2011. Springer Berlin Heidelberg. ISBN 978-3-642-23780-5.

[3] Riad Akrour, Marc Schoenauer, and Michèle Sebag. April: Active preference-learning based reinforcement learning, 2012. URL https://arxiv.org/abs/1208.0984.

[4] Chris L. Baker, Joshua B. Tenenbaum, and Rebecca Saxe. Goal inference as inverse planning. 2007. URL https://api.semanticscholar.org/CorpusID:1558263.

[5] Philip J. Ball, Laura Smith, Ilya Kostrikov, and Sergey Levine. Efficient online reinforcement learning with offline data, 2023. URL https://arxiv.org/abs/2302.02948.

[6] Daniel S. Brown, Wonjoon Goo, Prabhat Nagarajan, and Scott Niekum. Extrapolating beyond suboptimal demonstrations via inverse reinforcement learning from observations, 2019. URL https://arxiv.org/abs/1904.06387.

[7] Paul Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences, 2023. URL https://arxiv.org/abs/1706.03741.

[8] Benjamin Eysenbach, Tianjun Zhang, Ruslan Salakhutdinov, and Sergey Levine. Contrastive learning as goal-conditioned reinforcement learning, 2023. URL https://arxiv.org/abs/2206.07568.

[9] Johannes Fürnkranz, Eyke Hüllermeier, Weiwei Cheng, and Sang-Hyeun Park. Preference-based reinforcement learning: a formal framework and a policy iteration algorithm. *Mach. Learn.*, 89(1–2):123–156, October 2012. ISSN 0885-6125. doi: 10.1007/s10994-012-5313-8. URL https://doi.org/10.1007/s10994-012-5313-8.

[10] Dibya Ghosh, Abhishek Gupta, and Sergey Levine. Learning actionable representations with goal-conditioned policies, 2019. URL https://arxiv.org/abs/1811.07819.

[11] Abhishek Gupta, Vikash Kumar, Corey Lynch, Sergey Levine, and Karol Hausman. Relay policy learning: Solving long-horizon tasks via imitation and reinforcement learning, 2019. URL https://arxiv.org/abs/1910.11956.

[12] Abhishek Gupta, Aldo Pacchiano, Yuexiang Zhai, Sham M. Kakade, and Sergey Levine. Unpacking reward shaping: Understanding the benefits of reward engineering on sample complexity, 2022. URL https://arxiv.org/abs/2210.09579.

[13] Joey Hejna and Dorsa Sadigh. Few-shot preference learning for human-in-the-loop rl, 2022. URL https://arxiv.org/abs/2212.03363.

[14] Joey Hejna and Dorsa Sadigh. Inverse preference learning: Preference-based rl without a reward function, 2023. URL https://arxiv.org/abs/2305.15363.

[15] Joey Hejna, Rafael Rafailov, Harshit Sikchi, Chelsea Finn, Scott Niekum, W. Bradley Knox, and Dorsa Sadigh. Contrastive preference learning: Learning from human feedback without rl, 2024. URL https://arxiv.org/abs/2310.13639.

[16] Todd Hester, Matej Vecerik, Olivier Pietquin, Marc Lanctot, Tom Schaul, Bilal Piot, Dan Horgan, John Quan, Andrew Sendonaris, Gabriel Dulac-Arnold, Ian Osband, John Agapiou, Joel Z. Leibo, and Audrunas Gruslys. Deep q-learning from demonstrations, 2017. URL https://arxiv.org/abs/1704.03732.

[17] Borja Ibarz, Jan Leike, Tobias Pohlen, Geoffrey Irving, Shane Legg, and Dario Amodei. Reward learning from human preferences and demonstrations in atari, 2018. URL https://arxiv.org/abs/1811.06521.

[18] Leslie Pack Kaelbling. Learning to achieve goals. In *International Joint Conference on Artificial Intelligence*, 1993. URL https://api.semanticscholar.org/CorpusID:5538688.

[19] Sham Kakade and John Langford. Approximately optimal approximate reinforcement learning. In *Proceedings of the Nineteenth International Conference on Machine Learning*, ICML '02, page 267–274, San Francisco, CA, USA, 2002. Morgan Kaufmann Publishers Inc. ISBN 1558608737.

[20] Bingyi Kang, Zequn Jie, and Jiashi Feng. Policy optimization with demonstrations. In Jennifer Dy and Andreas Krause, editors, *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 2469–2478. PMLR, 10–15 Jul 2018. URL https://proceedings.mlr.press/v80/kang18a.html.

[21] Ilya Kostrikov, Ashvin Nair, and Sergey Levine. Offline reinforcement learning with implicit q-learning, 2021. URL https://arxiv.org/abs/2110.06169.

[22] Kimin Lee, Laura Smith, and Pieter Abbeel. Pebble: Feedback-efficient interactive reinforcement learning via relabeling experience and unsupervised pre-training, 2021. URL https://arxiv.org/abs/2106.05091.

[23] Seunghyun Lee, Younggyo Seo, Kimin Lee, Pieter Abbeel, and Jinwoo Shin. Offline-to-online reinforcement learning via balanced replay and pessimistic q-ensemble, 2021. URL https://arxiv.org/abs/2107.00591.

[24] Max Sobol Mark, Tian Gao, Georgia Gabriela Sampaio, Mohan Kumar Srirama, Archit Sharma, Chelsea Finn, and Aviral Kumar. Policy agnostic rl: Offline rl and online rl fine-tuning of any class and backbone, 2024. URL https://arxiv.org/abs/2412.06685.

[25] Ashvin Nair, Abhishek Gupta, Murtaza Dalal, and Sergey Levine. Awac: Accelerating online reinforcement learning with offline datasets, 2021. URL https://arxiv.org/abs/2006.09359.

[26] Ashvin V Nair, Vitchyr Pong, Murtaza Dalal,

Shikhar Bahl, Steven Lin, and Sergey Levine. Visual reinforcement learning with imagined goals. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/file/7ec69dd44416c46745f6edd947b470cd-Paper.pdf.

[27] Mitsuhiko Nakamoto, Yuexiang Zhai, Anikait Singh, Max Sobol Mark, Yi Ma, Chelsea Finn, Aviral Kumar, and Sergey Levine. Cal-ql: Calibrated offline rl pretraining for efficient online fine-tuning, 2024. URL https://arxiv.org/abs/2303.05479.

[28] Jongjin Park, Younggyo Seo, Jinwoo Shin, Honglak Lee, Pieter Abbeel, and Kimin Lee. Surf: Semi-supervised reward learning with data augmentation for feedback-efficient preference-based reinforcement learning, 2022. URL https://arxiv.org/abs/2203.10050.

[29] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model, 2024. URL https://arxiv.org/abs/2305.18290.

[30] Aravind Rajeswaran, Vikash Kumar, Abhishek Gupta, Giulia Vezzani, John Schulman, Emanuel Todorov, and Sergey Levine. Learning complex dexterous manipulation with deep reinforcement learning and demonstrations, 2018. URL https://arxiv.org/abs/1709.10087.

[31] Stefan Schaal. Learning from demonstration. In *Proceedings of the 10th International Conference on Neural Information Processing Systems*, NIPS'96, page 1040–1046, Cambridge, MA, USA, 1996. MIT Press.

[32] Tom Schaul, Daniel Horgan, Karol Gregor, and David Silver. Universal value function approximators. In Francis Bach and David Blei, editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1312–1320, Lille, France, 07–09 Jul 2015. PMLR. URL https://proceedings.mlr.press/v37/schaul15.html.

[33] Daniel Shin, Daniel S. Brown, and Anca D. Dragan. Offline preference-based apprenticeship learning, 2022. URL https://arxiv.org/abs/2107.09251.

[34] Yuda Song, Yifei Zhou, Ayush Sekhari, J. Andrew Bagnell, Akshay Krishnamurthy, and Wen Sun. Hybrid rl: Using both offline and online data can make rl efficient, 2023. URL https://arxiv.org/abs/2210.06718.

[35] Yuda Song, Gokul Swamy, Aarti Singh, J. Andrew Bagnell, and Wen Sun. The importance of online data: Understanding preference fine-tuning via coverage, 2024. URL https://arxiv.org/abs/2406.01462.

[36] Mel Vecerik, Todd Hester, Jonathan Scholz, Fumin Wang, Olivier Pietquin, Bilal Piot, Nicolas Heess, Thomas Rothörl, Thomas Lampe, and Martin Riedmiller. Leveraging demonstrations for deep reinforcement learning on robotics problems with sparse rewards, 2018. URL https://arxiv.org/abs/1707.08817.

[37] Yuexiang Zhai, Christina Baek, Zhengyuan Zhou, Jiantao Jiao, and Yi Ma. Computational benefits of intermediate rewards for goal-reaching policy learning, 2022. URL https://arxiv.org/abs/2107.03961.

[38] Henry Zhu, Abhishek Gupta, Aravind Rajeswaran, Sergey Levine, and Vikash Kumar. Dexterous manipulation with deep reinforcement learning: Efficient, general, and low-cost, 2018. URL https://arxiv.org/abs/1810.06045.

[39] Yuke Zhu, Ziyu Wang, Josh Merel, Andrei Rusu, Tom Erez, Serkan Cabi, Saran Tunyasuvunakool, János Kramár, Raia Hadsell, Nando de Freitas, and Nicolas Heess. Reinforcement and imitation learning for diverse visuomotor skills, 2018. URL https://arxiv.org/abs/1802.09564.

APPENDIX

Our hypothesis for Preference based CRL is that environment dynamics will inherently suppress label noise. This process of implicit arbitration is illustrated below.

A. Exploitation Signal (True Preferences)

Figure 9 illustrates the standard C-GCRL training process, where the agent's visited state distribution gradually shifts over time ($t \rightarrow t+n$) to match the goal distribution (p(sg)) through exploration and Bellman updates.

Figure 10 shows the benefit of adding our preference-alignment loss. The preference dataset provides regions of known positive states (s^+, green ellipse) and negative states (s^-, red ellipse) relative to the implicit goal g_{task}. Even before the agent's main state distribution (blue) has discovered the goal, the preference loss $\mathcal{L}_{pref}$ provides an explicit "push" or directional signal, early in the learning process. This serves as a form of exploitation, directing the policy to increase the Q-values of positive states and decrease those of negative states, effectively guiding exploration and accelerating learning.

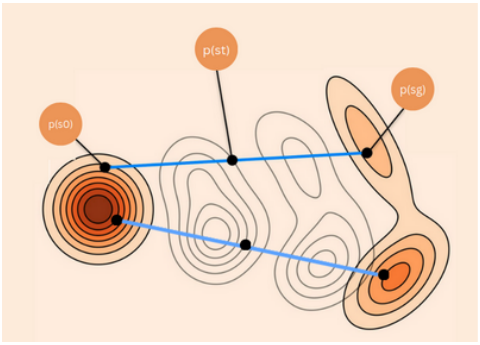


Fig. 9. State distribution at $t \rightarrow t+n$. Standard C-GCRL training

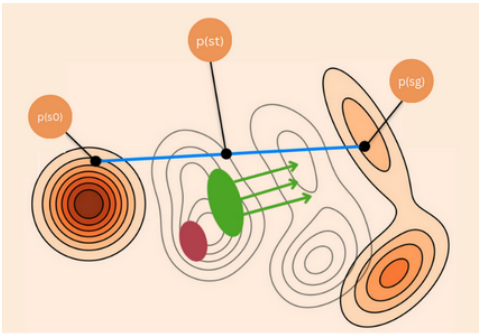


Fig. 10. Exploitation via preference signal. The preference signal (green/red) accelerates learning by providing an explicit exploitation signal towards the goal, independent of the agent's current exploration-driven state distribution.

B. Noise Suppression (False Preferences)

Figure 11 illustrates the key hypothesis for noise robustness.

- **At time** t**:** The noisy dataset D_n contains a true positive region (large green ellipse) and a false positive state (small green spot), which is incorrectly labeled as positive.
- **At time** $t+k$**:** As the policy learns, the C-GCRL objective (which relies on environment dynamics) discovers that trajectories originating from the region around the false positive do not lead to the goal. This dynamic signal creates a surrounding region of low value (red ellipse).
- **At time** $t+n$**:** The Bellman updates have propagated this low value, "pulling down" the goal conditioned Q-value of the false positive state. The signal from the environment dynamics has effectively overridden the incorrect signal from the single noisy label μ. The entire region is now correctly identified as negative (all red).

Our objective allows the C-GCRL signal, derived from environment dynamics, to implicitly counteract and override the noisy preference loss. It strengthens the signal from true preferences (which are consistent with the dynamics) and suppresses the signal from false positives (which conflict with the dynamics), leading to a robust policy.

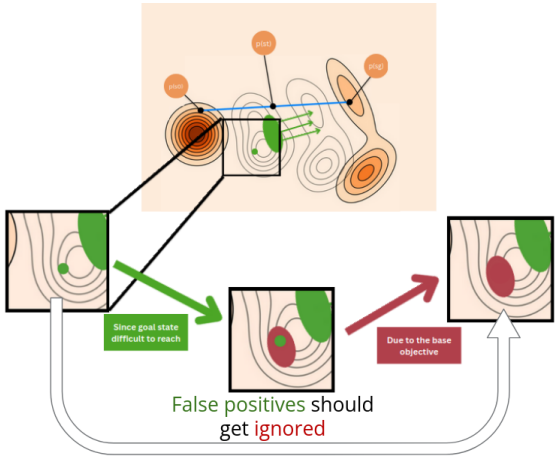


Fig. 11. Noise suppression at $t \rightarrow t+n$. Our method suppresses false positives by leveraging environment dynamics, eventually overriding the incorrect label (green spot) with the correct, low-value signal (red region).

C. Generalization of Preference Signals

Another benefit of this joint optimization stems from the C-GCRL objective itself. The contrastive loss is responsible for structuring the representation space such that states which are dynamically similar (i.e., from which the same goals can be reached) are mapped to nearby representations $\phi(s, a)$. The preference loss $\mathcal{L}_{pref}$ then operates on this already-structured space.

When a preference tuple ($s^+ \succ s^-$) induces a gradient update, it does not merely affect the Q-values of those two states in isolation. Because the C-GCRL objective enforces dynamic consistency, states s' that are dynamically similar to s^+ will already have representations $\phi(s', a')$ close to $\phi(s^+, a^+)$. Consequently, they will also inherit a similar "positive" update from the preference loss. Likewise, states dynamically similar to s^- will receive a "negative" update.

D. Experiments with Inverse preference learning

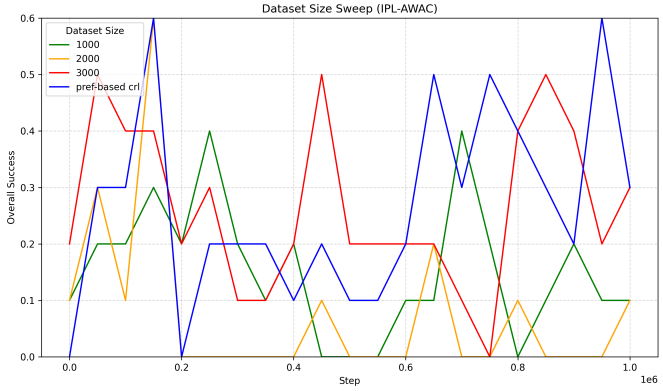


Fig. 12. Ablation with IPL with different dataset sizes under 0.3 noise level.

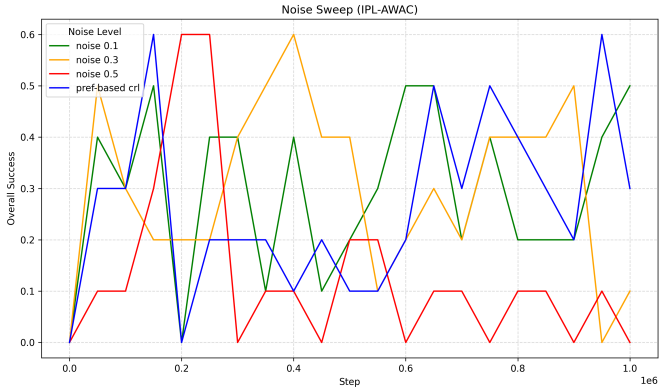


Fig. 13. Ablation with IPL on different noise levels.